DATA VISUALIZATION AND STORYTELLING

Task 2 | Sales Performance Analysis

Objective
Turn sales data into clear visual insights and practical business recommendations.

Tools
Power BI or Tableau concepts, chart design, business analysis, and GitHub documentation.

Deliverables
Charts, findings, recommendations, interview answers, and a submission checklist.

Important data note: No CSV dataset was attached. This report therefore uses clearly labeled illustrative sample values to demonstrate the charts and storytelling approach. Replace the example figures and findings with results from your actual Superstore.csv before claiming them as real analysis.

Beginner-friendly project report

1. Executive Summary

This project demonstrates how sales data can be turned into a visual story. It uses KPI cards, a monthly sales trend, category comparison, regional profit comparison, and sub-category profitability. The aim is to answer: How is the business performing? What drives the results? What action should management consider?

Illustrative headline metrics

$198,800	**$34,100**	**1,248**	**17.2%**

Sample Sales	Sample Profit	Sample Orders	Sample Profit Margin*

*All values are illustrative and are not verified results from a real Superstore file.

Story in one sentence

In this demonstration, sales peak in December, Technology leads category sales, West leads regional profit, and Tables shows a negative profit that should be investigated.

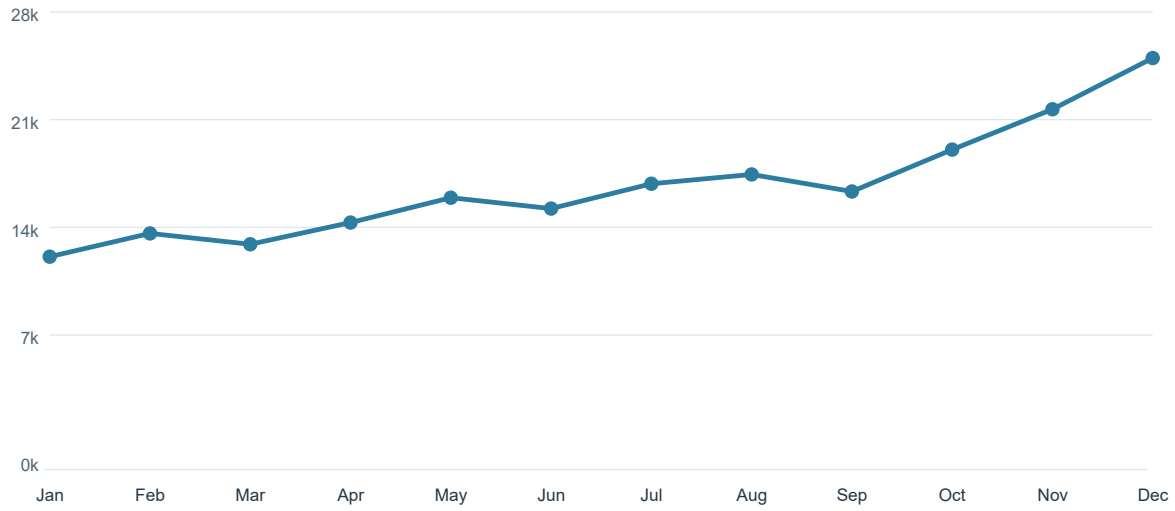
Business objective

Help managers make better decisions about product mix, promotions, inventory, and regional performance using clear, accurate and uncluttered visuals.

2. Monthly Sales Trend

Chart choice: A line chart is suitable for time-series data because it makes changes and direction across months easy to see.

Illustrative monthly sales



Finding: Sales rise from $12,000 in January to $24,800 in December in this example. December is the highest month ($24,800) and January is the lowest ($12,000).

Business interpretation

If the actual data shows a similar pattern, managers can prepare inventory, staffing and promotions ahead of peak periods. Check whether the trend is driven by more orders, larger orders or a particular category.

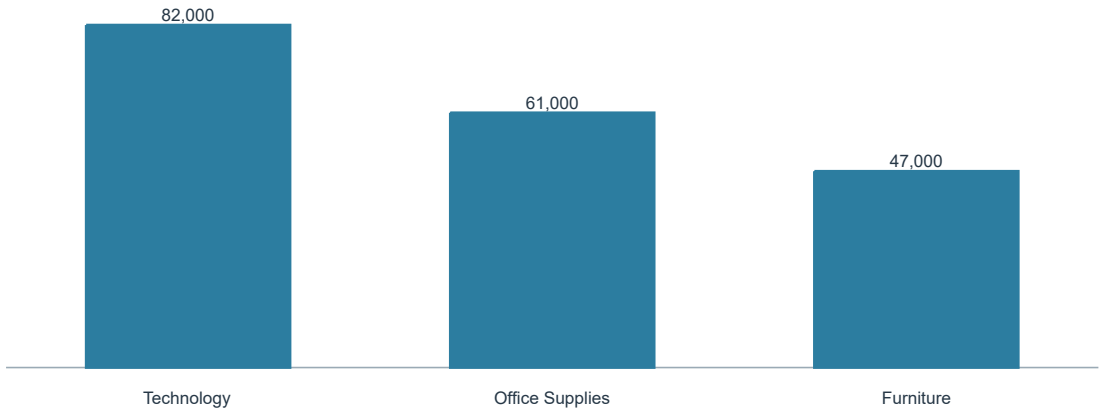
Recommendation

Compare the same months across multiple years before claiming seasonality. Add a year filter to support fair comparisons.

3. Sales by Product Category

Chart choice: A bar chart is effective for comparing values and ranking categories.

Illustrative sales by category



Finding: Technology has the highest illustrative sales at $82,000, followed by Office Supplies ($61,000) and Furniture ($47,000).

Business interpretation

High sales do not automatically mean high profit. Compare sales, profit and profit margin before increasing discounts or marketing spend.

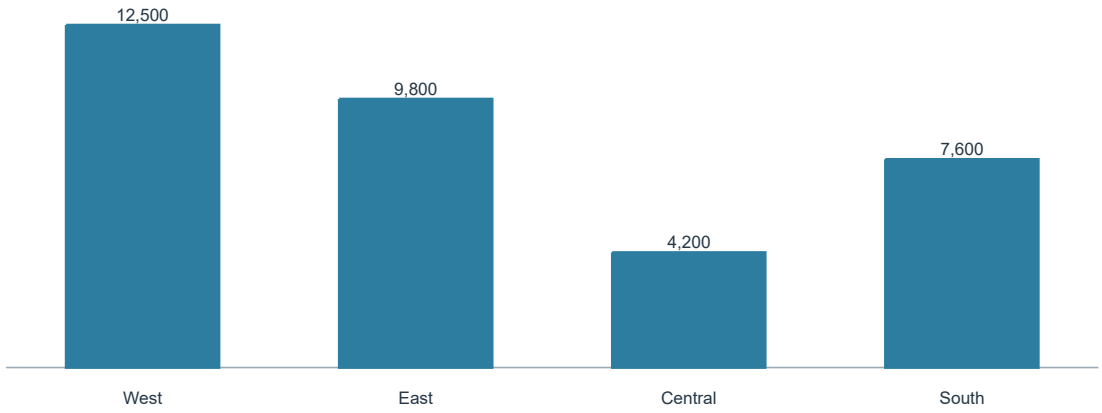
Recommendation

Focus on categories with strong sales and healthy margins. For weaker categories, investigate demand, pricing, stock availability and product mix before making changes.

4. Profit by Region

Chart choice: A bar chart makes regional profit differences straightforward to compare.

Illustrative profit by region



Finding: West has the highest illustrative profit ($12,500), while Central has the lowest ($4,200) among these four example regions.

Business interpretation

Lower profit may result from lower sales, heavier discounts, higher shipping costs or a different product mix. This chart shows where to investigate, but not the cause.

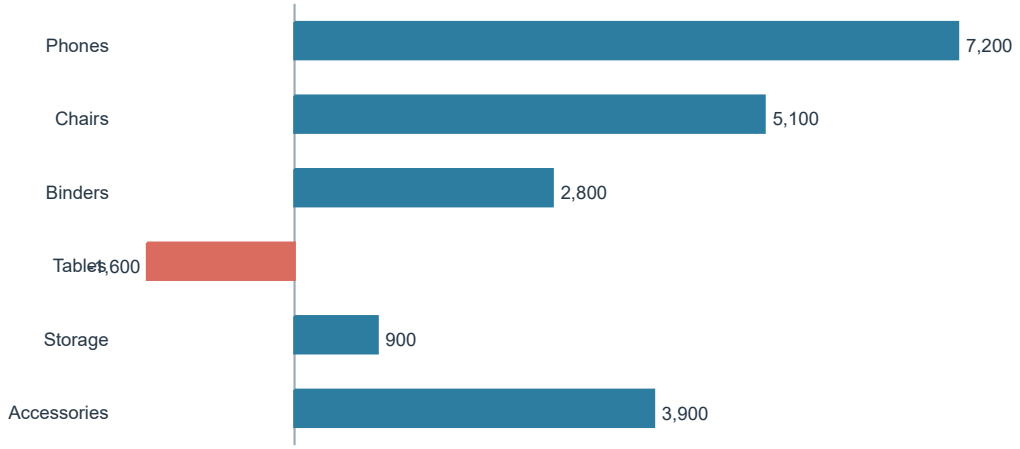
Recommendation

Break down the lowest-profit region by category, sub-category, discount and customer segment. Compare revenue and margin together.

5. Sub-Category Profitability

Chart choice: A horizontal bar chart supports longer labels and clearly displays both positive and negative profit.

Illustrative profit by sub-category



Finding: Tables has illustrative negative profit (-$1,600), while Phones has the highest illustrative profit ($7,200).

Business interpretation

A product may generate sales but still lose money when discounts, returns, shipping or costs are high. Negative profit is a signal to investigate, not an automatic reason to discontinue a product.

Recommendation

Review selling price, discount levels, product cost, shipping expense, returns and customer demand. Test changes and monitor their effect on profit.

6. Dashboard Design and Storytelling

Suggested dashboard layout

Area	Visual
Top row	KPI cards: Sales | Profit | Orders | Profit Margin
Middle left	Monthly sales trend — line chart

Middle right	Sales by category — bar chart
Bottom left	Profit by region — bar chart
Bottom right	Profit by sub-category — horizontal bar chart
Filters	Order Date, Region, Category, Customer Segment

Dashboard best practices

• Use one main message per chart.
• Keep colors consistent; use warning colors for negative or unusual values.
• Write meaningful chart titles and show units and date ranges.
• Avoid 3D effects, clutter, excessive labels and misleading axes.
• Add filters only when they help answer business questions.
• Place important KPIs near the top.

Story flow

1. Start with overall performance.
2. Show when sales change over time.
3. Compare categories and regions.
4. Find product-level profit issues.
5. End with recommendations.

7. Business Recommendations

Validate each recommendation against the real dataset before presenting it as a finding.

1. **Plan around demand** — Use monthly sales patterns to plan inventory, staffing and marketing ahead of peak periods.
2. **Focus on profitable growth** — Track profit and margin alongside sales so growth is not driven only by discounting.
3. **Investigate low-profit regions** — Break regional results down by category, discount, shipping and customer segment.
4. **Review loss-making products** — Examine costs, returns, discounts and pricing for sub-categories with negative profit.
5. **Monitor outcomes** — Refresh the dashboard regularly and check whether actions improve profit and margin.

Limitations and data quality

The sample values in this report are illustrative and were not calculated from an uploaded Superstore.csv file. Results may change with the real data and selected date range. Before final submission, check missing values, duplicate rows, data types, currency, distinct order IDs and negative profits. A chart can identify a pattern, but it does not prove the cause.

8. Interview Questions and Answers

1. What is the importance of data visualization?

It presents information through charts and graphs, helping people identify patterns, compare results, understand trends and make informed decisions quickly.

2. When do you use a pie chart versus a bar chart?

A pie chart is useful for showing a few categories as parts of a whole. A bar chart is better for comparing values across categories, especially when there are many categories or close values.

3. How do you make visualizations more engaging?

I use suitable charts, clear titles, consistent colors, readable labels, useful filters and short explanations of important findings. I avoid unnecessary decoration.

4. What is data storytelling?

Data storytelling combines data, visuals and explanation to communicate what happened, why it may have happened, and what action should be considered.

5. How do you avoid misleading visualizations?

I use accurate data, clear labels, appropriate scales and relevant comparisons. I avoid hiding data, exaggerating differences or claiming causation without evidence.

6. What are best practices in dashboard design?

Put key KPIs at the top, organize charts logically, use consistent colors, keep the dashboard uncluttered, provide meaningful filters, and show units and dates.

7. What tools have you used for visualization?

Describe the tool actually used. For example: I used Power BI Desktop to import sales data, create KPI cards and charts, add slicers and build a dashboard. Do not claim experience with a tool you have not used.

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